Charm production in muon deep-inelastic scattering with FASER*Cal*

Yields, charm/anticharm tagging, and Bjorken-x reconstruction

Internal note — generated from the `muondis-fasercal` analysis

August 5, 2026

Abstract

We study charm production in high-energy muon deep-inelastic scattering (DIS) in FASER*Cal*, using the POWHEG + Pythia8 muon-DIS samples developed for the FASERν programme, and the FASER*Cal* geometry and detector performance as given in the Conceptual Design Report and the July 2026 collaboration-meeting talk. Three results. First, the charm/anticharm asymmetry A_c — the observable that originally motivated the study — is *identically zero* in the available PDF sets, so it cannot be used as an intrinsic-charm observable without a PDF set carrying an explicit charm asymmetry. Second, the accessible intrinsic-charm signal is instead the large-x_B charm excess, and we confirm that the lepton-only x_B reconstruction used in the literature is destroyed by FASER*Cal*'s muon-momentum resolution. Third, and centrally, we show that *calorimetric* reconstruction methods — which FASERν cannot use — recover a factor ~ 3–3.8 of the large-x_B signal, and raise the intrinsic-charm purity of the selected sample from 64% to 95% when two statistically independent methods are combined. All uncertainties quoted are statistical and detector-resolution effects only; systematic uncertainties are discussed in Sec. 7 but are *not* included in any number.

Contents

1 Motivation and scope

The proton may carry a non-perturbative, "intrinsic" charm (IC) component: a $|uudc\bar{c}\rangle$ Fock state in which the charm quarks carry a large fraction of the proton momentum. Its signature is an excess of charm at large Bjorken $x_{\rm B}$ over what perturbative gluon splitting produces. High-energy muon DIS at the LHC is a clean probe, and FASER*Cal* — an electronic, highly segmented three-dimensional scintillator calorimeter with a downstream magnetised muon spectrometer — is a candidate detector.

This note addresses three questions:

1. How many muon-DIS charm events does FASER*Cal* collect, and can charm be distinguished from anticharm? (Sec. 3)
2. Is the charm/anticharm asymmetry A_c a usable intrinsic-charm observable? (Sec. 4)
3. Can $x_{\rm B}$ be reconstructed well enough at large $x_{\rm B}$ for the intrinsic-charm excess to survive? (Sec. 5)

The third is the critical one. Ref. [1] reconstructs $x_{\rm B}$ from the scattered muon alone and finds that a muon-momentum smearing of 10% already reduces the $x_{\rm B} \geq 0.2$ charm excess from $\times 8$ to $\times 2$, and that 30% removes it entirely. FASER*Cal*'s spectrometer is iron-core and multiple-scattering dominated, so this is a genuine threat — but FASER*Cal* is also a calorimeter, which FASERν is not.

2 Inputs and their provenance

Every number used is listed in Table 1 with its source. Where a quantity is an assumption rather than a documented value, this is stated explicitly; those are the entries that most need review.

Detector position and the off-axis flux. FASER*Cal* sits off the line of sight. The CDR/talk shift is $(452, 236)$ mm; the collaboration reports a further 12 cm, giving $(572, 236)$ mm. The FLUKA muon simulation (the same MC22 production that underlies the LHAPDF flux grid) evaluated over the 480×480 mm face at that point gives $F_{\rm flux} = 1.10$, with $\langle p_\mu \rangle = 1188$ GeV and a μ^+ fraction of 0.18.

The sign convention is *confirmed*: the FLUKA frame places FASER*Cal* at positive x, consistent with "left of the LoS looking downstream" in a right-handed frame. An earlier version

Quantity	Value	Source
Run 4 luminosity	$780\,\mathrm{fb}^{-1}$	FASER*Cal* talk; CDR Sec. 3.1
3DCal geometry	10 modules $\times$ 20 layers of $1\,\mathrm{cm}^3$ cubes	talk slide 4
3DCal transverse face	$48 \times 48\,\mathrm{cm}^2$	talk slide 11
Tungsten absorber	1, 5 or 10 mm *per module*	talk slide 39; CDR Table 5
3DCal total mass	581 / 896 / 1118 kg	CDR Table 5
Fiducial region	$z < 1150\,\mathrm{mm}$ (48% of length)	CDR Table 8
Position	LoS shift $(572, 236)$ mm, tilt 4.5°	talk + 12 cm (collab.)
Muon momentum resolution	20/22.8/29/63.1% at 20/100/200/1000 GeV	CDR Sec. 2.6.2
Charge misidentification	$< 0.7\%$ below 100 GeV; 2–5% at TeV	CDR Sec. 2.6.2
Spectrometer acceptance	43% (≥ 2 MDT stations)	talk slide 33
Hadronic energy resolution	$\sim 9\%$	talk (p_{jet}, NC)
Off-axis flux factor	$F_{\mathrm{flux}} = 1.10$	*this work*, from FLUKA
Run 4 optics enhancement	$F_{\mathrm{optics}} = 2.0$	CDR Sec. 1.3.2 (rate $\times 2$)
Muon angular resolution	0.3 mrad	*derived* from 3DCal geometry
Hadronic angular resolution	20 mrad	**ASSUMPTION – no source**
Identification + linking eff.	scanned 0.05–1	**not assumed; scanned**

Table 1: Inputs. Entries above the rule are taken from FASER*Cal* documents; below the rule are derived or assumed in this work. Where the talk and the CDR disagree, the talk (more recent) is used.

of this note flagged the opposite convention as an open factor-2.1 risk; that is now closed. The 12 cm shift itself matters little — across the four possible directions F_{flux} spans only 0.90–1.17.

2.1 Monte Carlo samples

Two POWHEG + Pythia8 productions differing *only* in the charm PDF:

Production	LHAPDF	Role
`production_v1`	331100, `NNPDF40_nnlo_as_01180`	fitted charm (IC allowed)
`production_pc`	332100, `NNPDF40_nnlo_pch_as_01180`	perturbative charm (null)

Muon beam on a fixed nucleon target, $Q_{\min} = 1.65\,\mathrm{GeV}$, four per-nucleon samples ($\mu^{\pm}$ on p/n). 25 independent seeds, 1.90×10^6 showered events per hypothesis. The muon flux is implemented as a beam PDF of a fictitious 7 TeV beam [1]; this has consequences discussed in Sec. 8.

Target composition is applied at analysis time: FASER*Cal* is polystyrene (CH, $w_p = 0.538$) plus tungsten ($w_p = 0.402$), not the tungsten-only target of FASERν.

3 The muon-DIS charm chain

3.1 Charm tagging without a vertex

In emulsion, charm is identified by imaging the sub-mm decay kink. In a scintillator this is impossible: the charm flight length is a few mm, at or below cell size, and in the interleaved-tungsten configuration the decay happens inside an absorber plate. **There is no displaced-vertex charm tag in FASER*Cal*.**

The surviving handle is the muon from the semileptonic charm decay, whose charge *is* the charm sign:

$$c \to s\,W^+,\ W^+ \to \mu^+\nu \qquad \bar{c} \to \bar{s}\,W^-,\ W^- \to \mu^-\bar{\nu}. \tag{1}$$

This gives an opposite-sign dimuon: the scattered DIS muon plus a softer muon of opposite charge. FASER*Cal* itself has no field; the sign comes from the downstream spectrometer. The identity $\mathrm{sign}(c) = q_\mu$ is verified event-by-event in the code and holds for 100% of events.

3.2 Yields

Table 2 gives the chain at Run 4 nominal conditions.

Stage	3DCal only	+AHCAL	Fraction
Muon DIS in fiducial volume	428 084	2 868 980	100%
containing charm	12 120	81 900	2.86%
charm $\to$ semileptonic μ	2 138	14 445	17.6% of charm
μ punches out of calorimeter	1 745	11 800	81.7%
reaches spectrometer (43%)	762	5 167	43%
identified + linked ($\varepsilon = 0.9$)	**686**	**4 650**	0.16% of DIS
Charge-tag purity	99.1%		CDR Sec. 2.6.2
Statistical reach $\sigma(A_c)$	0.038	0.015	

Table 2: Muon-DIS charm chain, Run 4 ($780\,\mathrm{fb}^{-1}$), 3DCal with 1 mm W per module, fiducial $z < 1150\,\mathrm{mm}$ (277 kg), $F_{\text{flux}} = 1.10$, $F_{\text{optics}} = 2.0$.

Two features dominate. The chain is **branching-limited, not detector-limited**: $2.86\% \times 17.6\% \approx 1/200$ before any detector effect, and no instrumentation improvement recovers it. And 94% of charm events contain *two* charm hadrons — the signature of photon-gluon fusion $\gamma g \to c\bar{c}$ — which is why the per-event semileptonic rate (17.6%) is about twice the per-hadron branching ratio.

Acceptance is an *angular* problem, not a momentum one: the decay muon has a median angle of 33 mrad against the scattered muon's 4.3 mrad, and 83% already pass the punch-through momentum threshold. The assumed 43% acceptance corresponds to a $\theta < 8.7\,\mathrm{mrad}$ cone. Charge identification is *not* a bottleneck: with a median decay-muon momentum of 11.5 GeV, the CDR misidentification is $< 0.7\%$.

3.3 Absorber options

Absorber	Fiducial mass	Tagged	$\sigma(A_c)$
1 mm W / module (baseline)	277 kg	686	0.038
5 mm W / module	421 kg	1032	0.031
10 mm W / module	514 kg	1262	0.028

3.4 Using the ECAL and AHCAL as active target volumes

The 3DCal is only 277 kg of fiducial target. The calorimeters behind it are far more massive, and — being sampling calorimeters — interactions in their *absorber* are measured too, so their whole mass counts as target.

Configuration	Fiducial mass	Muon DIS	Charm	Tagged	$\sigma(A_c)$
3DCal only	277 kg	428 084	12 120	686	0.038
3DCal + ECAL	1 047 kg	1 593 387	45 130	2 602	0.020
3DCal + AHCAL	1 874 kg	2 868 980	81 900	4 650	0.015
3DCal + ECAL + AHCAL	**2 644 kg**	**4 034 283**	**115 300**	**6 566**	**0.012**

Table 3: Adding the downstream calorimeters as target volumes. Masses are fiducial; all rows use $\varepsilon = 0.9$, which is *not* appropriate for the coarser detectors — see the caveat below.

Using the full stack gives a 9.5× larger target than the 3DCal alone and improves the statistical reach by ×3.2. Material composition is handled per volume: the ECAL is 92% lead ($w_p = 0.396$), the AHCAL 98% iron ($w_p = 0.464$), against polystyrene ($w_p = 0.538$) in the 3DCal.

Two caveats, and they are not small.

1. **Granularity.** The 3DCal has $1 \times 1\,\text{cm}^2$ transverse cells, the AHCAL $4 \times 4\,\text{cm}^2$, the ECAL $12 \times 12\,\text{cm}^2$ — a factor 144 between the extremes. Vertex finding and muon-to-vertex linking degrade accordingly, so **Table 3 should not be read at a common ε**: the ECAL and AHCAL rows belong at a substantially lower identification efficiency than the 3DCal row. This is precisely why the main result is presented *against* ε rather than at a fixed value.

2. **Containment.** The ECAL is only ~ 0.7 interaction lengths deep and cannot contain a hadronic shower by itself. It does however sit *upstream* of the AHCAL, which provides $\sim 5\lambda$ behind it, so it is given full fiducial acceptance here while the AHCAL — the last calorimeter — keeps the 48% of CDR Table 8. That asymmetry is a modelling choice, not a documented one.

The honest summary is that the extra mass is real and large, but it buys statistics in the detector volumes where the *tag* is weakest. Whether it is a net gain depends on how fast ε falls with cell size — a question only a full reconstruction can answer.

4 The charm/anticharm asymmetry is not a usable observable

4.1 It is zero by construction

Direct inspection of the LHAPDF grids gives, at $Q = 10\,\text{GeV}$:

x	$xc(x)$	$x\bar{c}(x)$	$(c-\bar{c})/(c+\bar{c})$
0.01	2.043×10^{-1}	2.047×10^{-1}	-1.0×10^{-3}
0.30	8.635×10^{-3}	8.669×10^{-3}	-2.0×10^{-3}
0.70	5.190×10^{-4}	5.191×10^{-4}	-7.3×10^{-5}

Over all 2352 grid points the maximum of $|c-\bar{c}|/(c+\bar{c})$ is 6.4×10^{-3}, and the number sum rule gives $\int(c-\bar{c})\,dx \,/ \int(c+\bar{c})\,dx = -1.5 \times 10^{-4}$. NNPDF4.0 fits the total $c+\bar{c}$ and sets $c-\bar{c}=0$. The residual is *larger* in the perturbative set than the fitted one — the signature of numerical noise rather than physics.

This is consistent with the light-cone argument: the minimal $|uudc\bar{c}\rangle$ Fock state contains one c and one $\bar{c}$ with identical distributions, so BHPS intrinsic charm predicts $A_c(x) = 0$ at every x. A non-zero A_c requires physics beyond the minimal Fock state, is model-dependent, and is constrained by $\int(c-\bar{c})dx = 0$.

4.2 But fragmentation generates one

Even with an exactly symmetric charm PDF, the *observable* sample is asymmetric, through the leading-baryon effect:

Species	$N(c)$	$N(\bar{c})$	Ratio	BR($\to \mu$)
Λ_c	13 430	1 963	**6.84**	4.8%
D^+	16 683	20 355	0.82	16.7%
D^0	31 730	38 820	0.82	6.6%

The c quark can pick up two valence quarks from the target remnant to form a Λ_c; the $\bar{c}$ cannot easily form an antibaryon. Λ_c also has the lowest semileptonic branching ratio of any charm hadron, so the c is preferentially *hidden from the muon tag* while the $\bar{c}$ is not, leaving a $\bar{c}$-enriched tagged sample: $A = -0.061 \pm 0.036$ ($N_{\text{eff}} = 778$, i.e. 1.7σ).

Consequences. A_c cannot be used as an IC observable with existing PDF inputs, and the fragmentation-induced asymmetry is of the *same size* as our statistical reach (0.041), so it is a competing background rather than a correction. Attention should move to the large-x_{B} excess.

4.3 The signal that does exist

The fitted/perturbative charm ratio at $Q = 10\,\text{GeV}$:

x	0.2	0.3	0.4	0.5	0.7
fitted / perturbative	1.8	3.2	5.7	10.9	44.3

At *event* level, integrated above $x_\text{B} = 0.2$, the charm sample is $\approx 99\%$ IC-attributable (Sec. 5.10).

5 Bjorken-x reconstruction

5.1 Kinematics and notation

For fixed-target DIS with nucleon mass M, incoming muon energy E_μ, scattered muon energy $E_{\mu'}$ and scattering angle θ_μ:

$$\nu = E_\mu - E_{\mu'}, \qquad Q^2 = 2E_\mu E_{\mu'}(1-\cos\theta_\mu), \qquad x_\text{B} = \frac{Q^2}{2M\nu}, \qquad y = \frac{\nu}{E_\mu}. \tag{2}$$

The hadronic system recoiling against the muon is exactly $X = q + p_\text{target}$, so it is fully determined by the muon kinematics:

$$E_\text{had} = \nu + M, \qquad p_T^\text{had} = E_{\mu'}\sin\theta_\mu, \qquad p_z^\text{had} = |p_\mu| - E_{\mu'}\cos\theta_\mu. \tag{3}$$

The key numbers of the problem, from our sample:

Median E_μ	665 GeV
Median $E_{\mu'}$	469 GeV
Median ν	52 GeV
Median y	0.161
Median θ_μ	4.3 mrad
Median θ_h	81 mrad

ν is only 7.7% of E_μ. It is a small difference of two large numbers, and this single fact drives everything that follows.

5.2 Method 1: lepton-only

What it does. Uses only the muon. Both muon energies and the scattering angle are measured; ν and Q^2 follow directly:

$$\nu^\text{lep} = E_\mu - E_{\mu'}, \qquad Q^2_\text{lep} = 2E_\mu E_{\mu'}(1-\cos\theta_\mu), \qquad x_\text{lep} = \frac{Q^2_\text{lep}}{2M\nu^\text{lep}}. \tag{4}$$

Why it is used. It is the only method available to an emulsion detector. FASERν measures angles superbly and energy not at all, so it is *forced* into this choice. It is what Ref. [1] uses.

Why it fails here. Because ν is a difference of two large numbers, a fractional error ϵ on each muon leg propagates as

$$\frac{\sigma_\nu}{\nu} \approx \frac{\epsilon}{y}\sqrt{1+(1-y)^2} \sim \frac{\epsilon}{y}. \tag{5}$$

With $y = 0.16$ and the CDR resolution $\sigma_p/p \approx 38\%$ at 469 GeV, this gives a *several-hundred-percent* error on ν. Measured directly, the ν resolution is an interquartile range of 410%. The migration matrix (Fig. 1) shows essentially *no* correlation between x_true and x_reco.

Additional problem. It requires E_μ. The downstream spectrometer sees only the *outgoing* muon, so if the incoming muon is not measured elsewhere this method **does not exist at all**.

5.3 Method 2: Jacquet–Blondel

What it does. Uses only the *hadronic* system. The energy transfer is measured directly by the calorimeter rather than inferred from a difference:

$$\nu^{\mathrm{JB}} = E_{\mathrm{had}} - M, \qquad Q^2_{\mathrm{JB}} = \frac{(p_T^{\mathrm{had}})^2}{1-y}, \qquad x_{\mathrm{JB}} = \frac{Q^2_{\mathrm{JB}}}{2M\nu^{\mathrm{JB}}}. \tag{6}$$

Why the target-mass subtraction matters. The measured hadronic energy is $\nu + M$ (Eq. 3); it includes the struck nucleon's rest energy. Omitting M biases x_{JB} low by M/ν, which is 9% on the $x_{\mathrm{B}} \geq 0.2$ sample where $\nu \approx 9\,\mathrm{GeV}$. This is a small but real effect that is easy to miss.

Strengths. The muon momentum is not used *at all*: a 9% hadronic resolution gives a 9% error on ν, not an amplified one. E_μ enters only through the mild $(1-y)$ factor, so the method survives when E_μ is unmeasured.

Weaknesses. Q^2 comes from the hadronic transverse momentum, which is the *difference* of a nearly-forward system — so it is sensitive to the hadronic *angular* resolution. In practice JB has excellent efficiency (78%) but poor purity (34%): the vastly more numerous low-x_{B} events migrate upward and flood the selection.

5.4 Method 3: the Σ method

The idea. Reconstruct the *incoming* muon energy from the final state, so that it need not be measured. With the beam along $+z$, the initial state has $E + p_z = 2E_\mu + M$, and since $E + p_z$ is conserved,

$$2E_\mu \;=\; \sum_{\mathrm{had}} (E + p_z) \;+\; E_{\mu'}(1 + \cos\theta_\mu) \;-\; M, \tag{7}$$

after which one proceeds exactly as in the lepton-only method but with this E_μ.

A necessary derivation. The collider Σ method uses $E - p_z$ [5]. For a *fixed target* with the beam along $+z$ that combination is identically M for the initial state and therefore carries no information. The correct fixed-target analogue uses $E + p_z$, as in Eq. 7. This was derived for this work.

Why it works — an exact cancellation. At the small angles relevant here, $E + p_z \approx 2E$, and Eq. 7 reduces to simple energy conservation, $E_\mu \approx E_{\mu'} + E_{\mathrm{had}} - M$. Then

$$\boxed{\nu^{\Sigma} \;=\; E_\mu^{\mathrm{reco}} - E_{\mu'}^{\mathrm{reco}} \;=\; \big(E_{\mu'} + E_{\mathrm{had}} - M\big) - E_{\mu'} \;=\; E_{\mathrm{had}} - M} \tag{8}$$

The muon-momentum error appears in both terms and cancels exactly. The resulting ν resolution is 13% (interquartile), against 410% for lepton-only where E_μ and $E_{\mu'}$ are measured independently and their errors *add*. This is an algebraic cancellation, not a tuned improvement, which is why the method is robust.

Insensitivity to the hadronic angle. $p_z = |p|\cos\theta_h$, and with $\theta_h \approx 81\,\mathrm{mrad}$ a $20\,\mathrm{mrad}$ smearing changes $\cos\theta_h$ by 0.2%. The method is therefore *first-order insensitive* to the hadronic angular resolution — the single largest unknown in the calorimetric methods.

Practical note. A user asking "can we measure E_μ by taking the spectrometer muon and adding the energy that went into the interaction?" is describing exactly this method. The two constructions are numerically identical here.

5.5 Method 4: double-angle

What it does. Uses only the two *angles*, plus the beam energy. Solving energy–momentum conservation for the scattered muon energy:

$$E_{\mu'}^{\mathrm{DA}} = E_\mu \frac{\sin\theta_h}{\sin(\theta_\mu + \theta_h)}, \tag{9}$$

then Q^2 and ν as usual. Being built from angles, it is immune to *energy-scale* errors in both the calorimeter and the spectrometer — a genuine advantage, since energy scales are the hardest thing to calibrate.

Weaknesses. Two.

1. It requires E_μ, and $x_{\mathrm{DA}} \propto E_\mu$ linearly. If the incoming muon is not measured, the method **does not exist**.
2. It depends entirely on the hadronic angle θ_h, which in a calorimeter is a *shower axis*, not a track. Its performance degrades steeply with σ_{θ_h} (Fig. 2, left): the figure of merit falls from 0.65 to 0.36 between 0 and 50 mrad.

Since σ_{θ_h} is currently an assumption with no documented value, the double-angle numbers carry the largest modelling uncertainty of the four.

5.6 Validation

With all resolutions set to zero, all four methods return $x_{\mathrm{reco}}/x_{\mathrm{true}} = 1.0000$. The framework is therefore verified *before* any smearing is applied. This test was essential: it exposed both the missing target-mass term in JB and a serious problem with the hadronic energy definition (Sec. 8).

5.7 Results

The figure of merit is efficiency × purity for the $x_{\mathrm{B}} \geq 0.2$ selection on charm events, where *efficiency* is the fraction of true $x_{\mathrm{B}} \geq 0.2$ events reconstructed above the cut, and *purity* is the fraction of selected events that truly have $x_{\mathrm{B}} \geq 0.2$.

Method	Efficiency	Purity	Eff.×Pur.	vs lepton	IC fraction
lepton-only	23.7%	63.8%	0.151	—	82%
Jacquet–Blondel	78.2%	33.6%	0.262	×1.7	38%
Σ	77.1%	57.6%	0.444	×2.9	64%
double-angle	82.5%	61.3%	0.506	×3.4	72%
Σ AND double-angle	66.3%	85.7%	**0.568**	×**3.8**	**95**%
all three	54.9%	94.7%	0.520	×3.4	97%

Table 4: Large-x_{B} signal retention, $x_{\mathrm{B}} \geq 0.2$, 25 seeds, all modelled effects included. "IC fraction" is the intrinsic-charm-attributable fraction of the selected sample, $1 - N_{\mathrm{pert}}/N_{\mathrm{fitted}}$.

5.8 Why combining works

The methods fail for *different* reasons, so their errors are not all correlated. Correlation of $\log(x_{\text{reco}}/x_{\text{true}})$:

	lepton	JB	Σ	DA
lepton-only	1.00	−0.03	0.65	−0.05
Jacquet–Blondel	−0.03	1.00	−0.01	**0.94**
Σ	0.65	−0.01	1.00	**0.08**
double-angle	−0.05	0.94	0.08	1.00

Σ and double-angle are **nearly independent** (0.08): one is driven by the calorimetric *energy*, the other by the *angles*. JB and double-angle are redundant (0.94), both being hadronic-system-driven. Requiring both members of the independent pair raises purity from 58% to 86% and the IC fraction of the selected sample from 64% to 95%.

Adding JB buys further purity but costs more efficiency than it gains. Note that this is a crude AND of two cuts; a proper constrained kinematic fit using all measurements with their covariances should do better still.

5.9 If E_μ is not measured — and the best method overall

The combination of Sec. 5.8 requires E_μ, because double-angle does. Since the downstream spectrometer sees only the *outgoing* muon, that may not be available. **The** 0.568 **quoted above therefore assumes a measured** E_μ, and this must be stated when it is used.

It turns out not to matter, because there is a better option that needs no incoming-muon measurement at all. Since $x_{\text{DA}} \propto E_\mu$, double-angle is only as good as its E_μ input — and the Σ estimator rebuilds E_μ from the final state using the $\sim$9% calorimeter rather than the $\sim$38% spectrometer. Feeding that into double-angle gives:

Method (E_μ *not* measured)	Efficiency	Purity	Eff.×Pur.
double-angle with Σ-reconstructed E_μ	81.8%	69.5%	**0.569**
Σ AND double-angle(Σ E_μ)	72.8%	73.6%	0.536
Σ AND Jacquet–Blondel	60.5%	87.5%	0.530
Σ alone	77.1%	57.6%	0.444
Jacquet–Blondel	78.4%	34.9%	0.274

Double-angle fed with the Σ-reconstructed E_μ is the best single method found, at 0.569 — marginally better than the two-method combination with a measured E_μ (0.568), and better than double-angle with a measured E_μ (0.506), for the same reason JB improved earlier: the calorimetric E_μ beats the spectrometer one. It requires no upstream measurement, so it is available in the configuration FASER*Cal* actually has. Combining it with Σ does *not* help — sharing the same reconstructed E_μ raises their error correlation from 0.08 to 0.19, and the AND then costs more efficiency than it buys purity.

All variants close at $x_{\text{reco}}/x_{\text{true}} = 1.0000$ with a perfect detector.

Recommendation. Use **double-angle with the Σ-reconstructed** E_μ. It is the best performer, needs no incoming-muon measurement, and avoids the two-cut combination entirely.

5.10 Intrinsic-charm content of the selection

At truth level the $x_{\text{B}} \geq 0.2$ cut is well placed:

x bin	0.1–0.2	0.2–0.3	0.3–0.4	> 0.4
IC fraction	85%	98%	99%	$\sim$ 100%

Integrated above 0.2 it is $\approx 99\%$. What is actually *selected* is diluted by migration from low x_{B}, where there is no IC — this is the purity restated in physics terms, and it is why the method combination matters (Table 4, last column).

Caveat: the perturbative sample above $x_{\mathrm{B}} = 0.2$ has $N_{\mathrm{eff}} = 7$ (966 raw events, one carrying 9% of the weight). The 99% survives only because the perturbative contribution is small to begin with; the perturbative *yield* itself must not be quoted, and any ratio-based statement there needs a dedicated high-statistics `production_pc`.

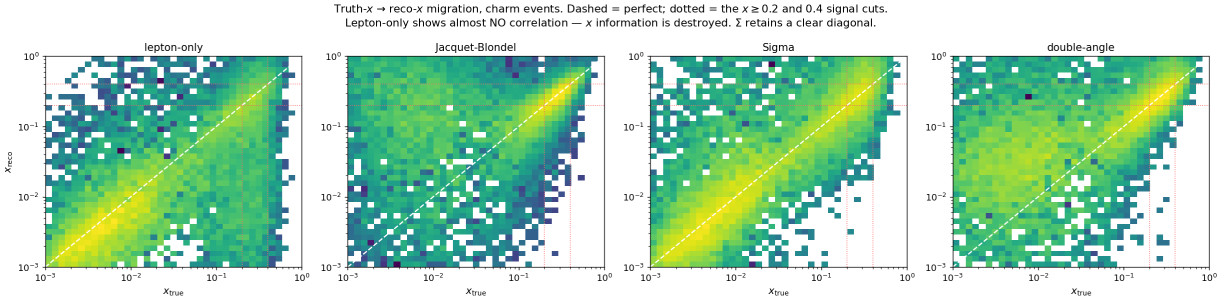

Figure 1: Truth-x_{B} $\rightarrow$ reco-x_{B} migration for charm events. Dashed line is perfect reconstruction. Lepton-only shows almost no correlation; Σ retains a clear diagonal.

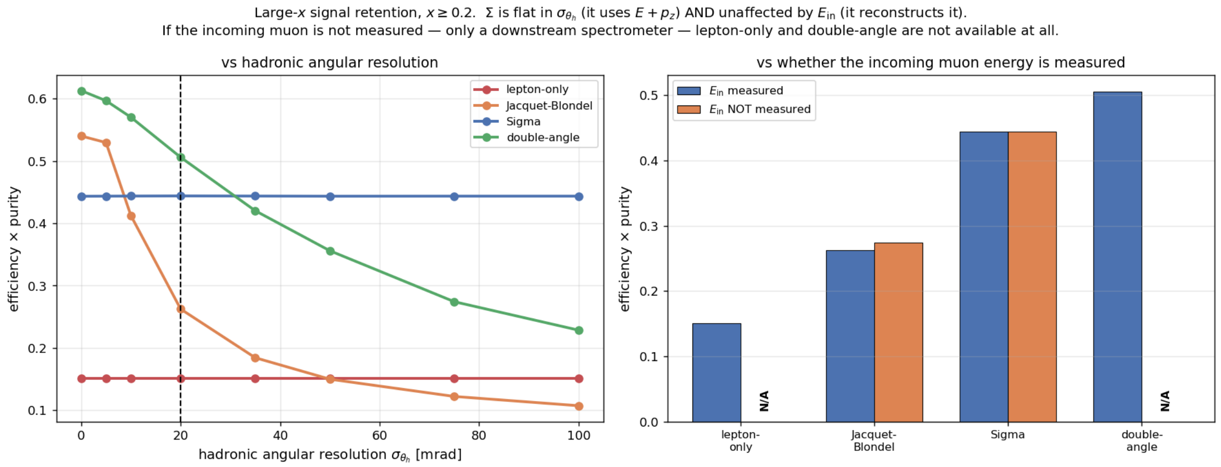

Figure 2: Left: signal retention versus the hadronic angular resolution. Σ is flat; double-angle degrades steeply. Right: versus whether the incoming muon energy is measured — lepton-only and double-angle do not exist without it.

6 Is $A_c(x)$ measurable at large x? No — and not for statistical reasons

Section 4 showed that A_c integrated over x is zero in the available PDF sets. That is however not the end of the theoretical story: *meson-cloud* models, in which the proton fluctuates to $|\Lambda_c^+ \bar{D}^0\rangle$ rather than to a minimal five-quark state, do predict $c(x) \neq \bar{c}(x)$. The c ends up in the baryon and the $\bar{c}$ in the meson; their different masses give different momentum fractions. This is the mechanism behind the recently proposed D^+/D^- asymmetry observable [4].

The predicted signal. Using the Pumplin–Lai–Tung meson-cloudy-bag parametrisation employed in Ref. [4], $f_c(x) \propto x^{1.897}(1-x)^{6.095}$ and $f_{\bar{c}}(x) \propto x^{2.5}(1-x)^{4.929}$, normalised by the quark-number sum rule, and diluting by the symmetric perturbative charm using our own fitted/perturbative ratios:

x	0.25	0.35	0.45	0.55	0.70
$A_{\mathrm{IC}}(x)$	$+0.129$	-0.055	-0.224	-0.384	-0.613
IC fraction	0.58	0.76	0.87	0.93	0.98
observable $A(x)$	$+0.075$	-0.042	-0.196	-0.359	-0.600

The asymmetry **changes sign at** $x = 0.318$ and reaches -0.6 at $x = 0.7$. Two things follow immediately. First, this *explains* the null result of Sec. 4 rather than contradicting it: integrated over x the asymmetry cancels by construction (the sum rule $\int(c-\bar{c})dx = 0$), so an inclusive $A_c \approx 0$ is exactly what the theory predicts. Second, at $x \gtrsim 0.5$ the predicted signal (-0.36 to -0.60) is 5–10 times larger than the fragmentation-induced background of Sec. 4 (-0.06), so the signal-to-background in the large-x bins would be favourable.

But the tag and the signal region are kinematically incompatible. We therefore evaluated the tagged yield differentially in x. The result is unambiguous and does not depend on any statistical argument:

True x	Charm events	with semilept. μ	punch through	**accepted**	acceptance
0.00–0.05	140 057	25 483	21 929	6 169	4.40%
0.05–0.10	10 530	1 797	1 148	38	0.36%
0.10–0.20	6 942	1 140	558	5	0.07%
0.20–0.32	4 583	726	285	1	0.02%
0.32–0.50	2 684	401	165	**0**	**0.00%**
0.50–1.00	299	51	28	**0**	**0.00%**

Table 5: Raw Monte Carlo counts through the charm tag, versus true x_{B} (25 seeds, 1.9×10^6 events). Of 6 213 events passing the full tag, **not one** has $x_{\mathrm{B}} \geq 0.32$.

The mechanism is simple. Large x_{B} means small ν at fixed Q^2, so the hadronic system — and hence the charm — is produced with little energy. The semileptonic decay muon is correspondingly soft and wide-angle:

	$x_{\mathrm{B}} < 0.1$	$x_{\mathrm{B}} \geq 0.32$
decay-muon median p	13.1 GeV	1.9 GeV
decay-muon median θ	29 mrad	257 mrad
spectrometer acceptance cone: 9.9 mrad		

A 257 mrad muon cannot enter a 9.9 mrad cone. **The dimuon charm charge-tag and the large-x_{B} signal region are mutually exclusive.**

Consequence. A dedicated production with an artificially asymmetric charm PDF was considered and **not** performed: it would change the generated $c/\bar{c}$ content but not the acceptance, so the tagged yield above $x = 0.32$ would remain zero. The handful of events reconstructed at large x_{reco} (5–9 raw MC events per bin) are migrated low-x events — their median *true* x_{B} is 0.01 — carrying no asymmetry.

What this does and does not kill.

- $A_c(x)$ **at large** x **is not measurable** with the dimuon tag in this configuration. Measuring it requires a charm tag that does not demand the decay muon in the spectrometer.
- **The large-x charm *rate* measurement is unaffected.** It needs no charge information, hence no decay muon in the spectrometer, and it is the subject of Sec. 5. That result stands.

- The natural alternative for large-x charm is the **charm-to-hadrons** channel, which the FASER*Cal* ML tagger already addresses (21% efficiency at 82% purity for $c \to H$ in neutrino events). It gives no charge information — so it can measure the large-x excess but not the asymmetry.

7 Systematic uncertainties — NOT included

Every number in this note is a statistical or detector-resolution effect. No systematic uncertainty is included anywhere. This section lists what is missing, with an estimate where one can be made cheaply.

7.1 What was varied, and what it does

Variation	Σ	double-angle	lepton-only
nominal	0.444	0.506	0.151
hadronic resolution 9% → 15%	0.442	0.506	0.151
hadronic resolution 9% → 5%	0.444	0.506	0.151
muon resolution ×1.3	0.378	0.456	0.131
muon resolution ×0.8	0.521	0.561	0.175
hadronic angle 20 → 50 mrad	0.444	0.356	0.151
no neutrino subtraction	0.433	0.506	0.151

Table 6: Sensitivity of the figure of merit to the modelled resolutions.

The dominant sensitivity is to the **muon momentum resolution**, even for the calorimetric methods — because although ν is calorimetric, Q^2 still uses the muon. The hadronic *energy* resolution is essentially irrelevant over 5–15%. The hadronic *angular* resolution matters only for double-angle.

7.2 An actionable consequence: tracker granularity

The CDR quotes the effect of finer SciFi granularity. Scaling to our momentum:

SciFi pitch	σ_p/p at 469 GeV	Σ	double-angle	Σ AND DA
100 μm (baseline)	38%	0.444	0.506	0.568
70 μm	29%	0.538	0.572	0.635
50 μm	26%	0.570	0.605	0.658

Going from 100 to 50 μm improves the combined selection by 16%. Since the muon resolution is the dominant lever, this is the single most effective detector change for this measurement.

7.3 Systematics not evaluated at all

1. **PDF and QCD-scale uncertainties.** The LHE files carry `rep001--rep103` NNPDF4.0 replica and scale weights, but only the nominal weight is read; the replicas are discarded at both the parsing and showering stages. *No PDF error band exists anywhere in this note.* For the fitted-vs-perturbative comparison this matters most, since the charm PDF at large x is precisely the poorly-known quantity.

2. **Absolute flux normalisation.** A residual factor ~ 1.7 discrepancy with Ref. [1] remains unexplained after correcting the reference target mass. $F_{\text{optics}} = 2.0$ is the CDR's own estimate for a 2024 configuration, not Run 4, and the CDR states it is unclear whether the increase can be mitigated. $F_{\text{flux}} = 0.96$ assumes the FLUKA x convention matches the FASER*Cal* one.

3. **Energy scale.** Only resolutions were varied, never a *bias*. A hadronic energy scale offset propagates directly into ν and hence x_{B}, and on a steeply falling spectrum a few-percent scale error can move the $x_{\mathrm{B}} \geq 0.2$ yield substantially. This should be evaluated.

4. **Hadronisation model.** The $\Lambda_c/\bar{\Lambda}_c = 6.8$ leading-baryon effect, which drives the fragmentation asymmetry of Sec. 4, is Pythia-specific and is not varied.

5. **Nuclear effects.** Free-nucleon PDFs are used; no nuclear modification is applied for carbon or tungsten.

6. **Detector modelling.** No reconstruction simulation: no voxel clustering, no shower overlap, no pile-up from the $\sim 1\,\mathrm{Hz/cm^2}$ muon background, and no π/K decay-in-flight background to the soft second muon.

7. **The hadronic angular resolution** $\sigma_{\theta_h} = 20\,\mathrm{mrad}$ has no documented source.

7.4 Statistical limitations

The perturbative sample above $x_{\mathrm{B}} = 0.2$ has $N_{\mathrm{eff}} = 7$. Any statement involving the perturbative *yield* or a fitted/perturbative *ratio* above $x_{\mathrm{B}} = 0.2$ is statistics-limited. The efficiency and purity numbers, which use only the fitted sample (7566 raw events above the cut, $N_{\mathrm{eff}} = 5319$), are not affected.

8 A methodological problem worth recording

The hadronic four-vector is computed from momentum conservation (Eq. 3), *not* by summing Pythia final-state particles. The particle sum is contaminated: the muon flux is implemented as a beam PDF of a fictitious 7 TeV beam, and that beam's remnant deposits a few GeV of hadronic activity that is not part of the DIS vertex. Energy conservation $E_\mu + M - E_{\mu'} - E_{\mathrm{had}} \geq 0$ fails in 80% of events, rising to 98% on the $x_{\mathrm{B}} \geq 0.2$ charm sample where E_{had}/ν reaches 2.5. It is negligible at large ν ($E_{\mathrm{had}}/\nu = 1.02$ above 500 GeV) but dominates exactly where the physics is.

Three particle-level fixes were tried and all failed: dropping beam-remnant particles (their hadronic *descendants* remain); a forward-cone cut (moves the mean, not the median); and requiring mother-chain ancestry to reach the target beam (*worse*, $E_{\mathrm{had}}/\nu = 8.5$, because Pythia's colour reconnection leaves the ancestry graph too connected to separate the two beam sides). Using conservation is exact and makes the closure exact by construction.

The cost is that neutrino losses must be added back explicitly, which is done: neutrinos from charm semileptonic decays are nonzero in 35% of charm events and carry a median 13.2 GeV (8.4% of ν) when present. Their effect is -10% on JB and negligible on Σ.

9 Summary

Next steps

1. Evaluate PDF and scale uncertainties using the `rep001--rep103` weights, which requires carrying them through the showering step.

2. Evaluate energy-*scale* systematics, not just resolutions.

3. Replace $\sigma_{\theta_h} = 20\,\mathrm{mrad}$ with a value from the FASER*Cal* reconstruction.

4. Implement a constrained kinematic fit (cf. Ref. [1] Sec. 4.2) in place of the crude AND of two cuts.

Question	Answer
How many tagged charm events?	686 charge-signed, vertex-linked charm muons in the 3DCal alone at 780 fb^{-1}; 2602 adding the ECAL, 4650 adding the AHCAL, 6566 using all three ($\sigma(A_c)$ improving 0.038 $\to$ 0.012). The extra volumes are coarser, so those rows belong at lower ε (Sec. 3.4).
Is charm/anticharm separable?	Yes in principle — the decay-muon charge *is* the charm sign, and the CDR spectrometer misidentifies $< 0.7\%$ — but the reach is $\sigma(A_c) = 0.041$.
Is A_c an intrinsic-charm observable?	**No.** It is identically zero in both available PDF sets. A non-zero A_c needs a PDF set with explicit charm asymmetry, or a model. A fragmentation-induced asymmetry of the same size as our reach competes with it.
What is the accessible IC signal?	The large-$x_{\rm B}$ charm excess: $\times 1.8$ at $x_{\rm B} = 0.2$ rising to $\times 44$ at 0.7. Above $x_{\rm B} = 0.2$ the charm is $\approx 99\%$ IC-attributable.
Does muon smearing destroy it?	For lepton-only, **yes** — ν is 7.7% of E_μ, so 38% muon resolution gives a several-hundred-percent ν error and no correlation between $x_{\rm true}$ and $x_{\rm reco}$.
Do calorimetric methods rescue it?	**Yes.** Σ retains $\times 2.9$ more signal; Σ AND double-angle $\times 3.8$ *if E_μ is measured*. If it is not — the likely case — double-angle fed with the Σ-reconstructed E_μ gives 0.569, $\times 3.8$, with no upstream measurement (Sec. 5.9).
Which method should be used?	**Double-angle with the Σ-reconstructed E_μ** (0.569): best performer, needs no incoming-muon measurement. Σ itself remains the robust fallback — flat in the hadronic angular resolution and unaffected by the neutrino loss.
Do neutral particles undermine it?	No. Neutrinos cost JB 10% and Σ nothing; neutral hadrons are 8% of the visible energy.
What most improves the measurement?	The **muon momentum resolution**. Going from 100 to 50 μm SciFi pitch improves the combined selection by 16%.
Is $A_c(x)$ measurable at large x?	**No**, and for a kinematic not a statistical reason: of 6213 tagged MC events, none has $x_{\rm B} \geq 0.32$. Large $x_{\rm B}$ implies small ν, hence a soft (1.9 GeV) wide-angle (257 mrad) decay muon that cannot enter the 9.9 mrad spectrometer cone. The tag and the signal region are mutually exclusive (Sec. 6).
Are systematics included?	**No. Nothing in this note includes a systematic uncertainty** (Sec. 7). PDF uncertainties in particular are entirely absent, although the replica weights exist in the LHE files.

Table 7: Summary of findings.

5. Generate a high-statistics perturbative-charm sample so that fitted/perturbative ratios above $x_{\mathrm{B}} = 0.2$ become meaningful.

6. Simulate the π/K decay-in-flight background to the soft second muon.

References

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[4] G. I. Lykasov, A. A. Sorokovikov and S. J. Brodsky, "Intrinsic charm and the D^+–D^- asymmetry produced in proton-proton collisions", Phys. Rev. D **111** (2025) 074035, arXiv:2501.02507. Parametrisation originally from J. Pumplin, H. L. Lai and W. K. Tung, Phys. Rev. D **75** (2007) 054029.

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